

Asterisms



162
generated patterns

9,036
generated stars

79
passing tests

SYSTEM SUMMARY

Licensed astronomy sources	->	Sync, build, validate pipeline	->	Compact generated catalogue
Browser gameplay	<->	Next.js API routes	<->	Restricted runtime database access

KEY ENGINEERING DECISIONS

- Joined multiple astronomy catalogues so puzzle shapes use real coordinates, magnitudes, and identifiers instead of hand-positioned points.
- Projected RA/Dec with a gnomonic tangent-plane projection, then rotated and scaled patterns without distorting geometry.
- Made Daily deterministic from UTC date so the puzzle remains playable even if score persistence is unavailable.
- Separated realistic star rendering from touch-friendly SVG hit targets.

LIMITATIONS

- Not a professional astronomy simulator or location-based observing planner.
- No accounts; score identity is anonymous and device-local.
- Score validation is a plausibility filter, not full anti-cheat.
- Mythology and broader sky-culture story text are deferred pending reliable sources.

QUALITY + RELIABILITY

- Current validation passes: data validation, tests, typecheck, lint, and production build.
- Data validation reports zero warnings and zero hard failures.
- Database-backed routes degrade gracefully; core gameplay catalogue is static app data.
- CI runs validation, tests, typecheck, lint, and build.

WHAT IT DEMONSTRATES

Asterisms demonstrates domain learning, data pipeline design, interactive rendering, product UX, API design, database-backed leaderboards, graceful degradation, and validation discipline across a real full-stack project.

Public-safe case study. No app source code, credentials, private repository links, account identifiers, or internal project IDs included.